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pQuant: Decoupled Linear Quantization-Aware Training

- □□: Wenzheng Zhang, Bingzheng Liu, Yang Hu □
- **arXiv**: 2602.23351
- □□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
- □□: □ scratch □□ QAT □□□□

NanoQuant: Efficient Sub-1-Bit Quantization

- □□: Hyochan Chong, Dongkyu Kim, Changdong Kim, Minseop Choi
- **arXiv**: □□□ 2026□2□6□
- □□: □□ LLM □□1□□□□□□□□□□
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Quecto-V1: 8-bit Quantized Small Language Models

- □□: Subrit Dikshit
- **arXiv**: □□□ 2026□2□18□
- □□: 8□□□□□□□□□□□□□□□□□□□□□□□□
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TernaryLM: Native 1-Bit Quantization

- □□: Nisharg Nargund, Priyesh Shukla

- **arXiv**: □□□ 2026□2□7□
- □□: □□□□□□□□□□□□1□□□□□□□□□□□□□
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RaBiT: Residual-Aware Binarization Training

- □□: Youngcheon You, Banseok Lee, Minseop Choi □
- **arXiv**: □□□ 2026□2□5□
- □□: □□□□□□ LLM □□□□□□□□□
- □□: □□□(±1)□ + □□□□□ matmul-free □□

BPDQ: Bit-Plane Decomposition Quantization

- □□: Junyu Chen, Jungang Li, Jing Xiong □
- **arXiv**: □□□ 2026□2□3□
- □□: □□ LLM □□□□□□□□□□□□□
- □□: □□□□□□□□□□

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QTALE: Quantization-Robust Token-Adaptive Layer Execution

- □□: Kanghyun Noh, Jinheon Choi, Yulhwa Kim
- **arXiv**: 2602.22207
- □□: □□□□□ Token □□□□□□□□
- □□: □□□□□□□□□□

Astro: Activation-guided Structured Regularization

- □□: Xi Chen, Ming Li, Junxi Li □
- **arXiv**: □□□ 2026□2□7□
- □□: □□□□□□□ LLM □□□□□□□□□□□□□□□
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D^{2}Quant: Accurate Low-bit Post-Training Weight Quantization

- □□: Xianglong Yan, ChengZhu Bao □
- **arXiv**: □□□ 2026□2□3□
- □□: □□□□□□□□□□□□

MatGPTQ: Matryoshka Quantization

- □□: Maximilian Kleinegger, Elvir Crnčević, Dan Alistarh
- **arXiv**: □□□ 2026□2□3□

- □□: □□□□□ Matryoshka □□□□□
- □□: □□□□□□□

MoBiQuant: Mixture-of-Bits Quantization

- □□: Dongwei Wang, Jinhee Kim, Seokho Han □
- **arXiv**: □□□ 2026□2□21□
- □□: Token □□□□□ LLM □□□□□□
- □□: □□ token □□□□□□

Regularized Calibration with Successive Rounding

- □□: Seohyeon Cha, Huancheng Chen, Dongjun Kim □
- **arXiv**: □□□ 2026□2□5□
- □□: □□□□□□□□□□□□□□□□□

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Quantization-Robust LLM Unlearning via LoRA

- □□: João Vitor Boer Abitante, Joana Meneguzzo Pasquali □
- **arXiv**: □□□ 2026□2□13□
- □□: □□□□□□□□□□□□ LLM □□□□
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Differential Privacy + Split Inference

- □□: Yujie Gu, Richeng Jin, Xiaoyu Ji □
- **arXiv**: □□□ 2026□2□11□
- □□: □□□□□□□□□ LLM □□□□
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TOM: Ternary Read-only Memory Accelerator

- □□: Hongyi Guan, Yijia Zhang, Wenqiang Wang □
- **arXiv**: □□□ 2026□2□24□
- □□: LLM □□□□□□□□□□□□□□
- □□□□: □□□□□□

Compact LLM Deployment in Mobile Edge Computing

- □□: Ruichen Zhang, Xiaofeng Luo, Jiayi He □
- **arXiv**: □□□ 2026□2□14□
- □□: □□□□□□□□□□ LLM □□□□□□□□
- □□: □□□□ + □□□□

Mapping Gemma3 onto Edge Dataflow Architecture

- □□: Shouyu Du, Miaoxiang Yu, Zhenyu Xu □
- **arXiv**: □□□ 2026□2□24□
- □□: Gemma3 □ AMD Ryzen AI NPU □□□□□
- □□: □□□□□□FlowQKV

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Scaling State-Space Models with Tensor Parallelism

- □□: Anurag Dutt, Nimit Shah, Hazem Masarani, Anshul Gandhi
- **arXiv**: □□□ 2026□2□24□
- □□: □ GPU □□□□□□□□□□□□
- □□: □□□ GPU □□/□□□□

SpeContext: Speculative Context Sparsity

- □□: Jiaming Xu, Jiayi Pan, Hanzhen Wang □
- **arXiv**: □□□ 2025□11□29□
- □□: □□□□□□□□□□□
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UniComp: Unified Evaluation of LLM Compression

- **arXiv**: □□□ 2026□2□11□
- □□: □□□□□□□□□ LLM □□□□□
- □□: □□□□□

SPQ: SVD-Pruning-Quantization Envelope

- **arXiv**: □□□ 2026□2□20□
- □□: □□□□□□□□□□□□

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GradPruner: Gradient-Guided Layer Pruning

- □□: Wei Huang, Anda Cheng, Yinggui Wang
- **arXiv**: □□□ 2026□1□27□
- □□: □□□□□□□□□□□ LLM □□□□□□□

FineScope: Precision Pruning with SAE

- **arXiv**: □□□ 2025□
- □□: □□ SAE □□□□□□□□□□□□ LLM □□□□

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SlimMoE: Expert Slimming and Distillation

- □□: Zichong Li, Chen Liang, Zixuan Zhang □
- **arXiv**: □□□ 2025□6□23□
- □□: □□□□□□□□□□□□ MoE □□□□□□□□□

daDPO: Distribution-Aware DPO

- □□: Zhengze Zhang, Shiqi Wang, Yiqun Shen □
- **arXiv**: □□□ 2025□6□2□
- □□: □□□□□□□□□□□□□ DPO

EPiC: Edge-Preserving CoT Condensation

- □□: Jinghan Jia, Hadi Reisizadeh, Chongyu Fan □
- **arXiv**: □□□ 2025□6□4□
- □□: □□□□□□□□□□□□□□□□□□□□□□

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QVLA: Vision-Language-Action Model Quantization

- □□: Yuhao Xu, Yantai Yang, Zhenyang Fan □
- **arXiv**: □□□ 2026□2□3□
- □□: VLA □□□□□□□□□□□□□□□□

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AdaTSQ: Temporal-Sensitivity Quantization for DiT

- □□: Shaoqiu Zhang, Zizhong Ding, Kaicheng Yang □
- **arXiv**: □□□ 2026□2□10□
- □□: Diffusion Transformer □□□□□□□
- □□: □□□□/□□□□

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Price of Universality in Vector Quantization

- □□: Alina Harbuzova, Or Ordentlich, Yury Polyanskiy
- **arXiv**: □□□ 2026□2□5□
- □□: □□□□□□□□□□□□□ 0.11 □□
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Rethinking Perplexity: Input Length Impact

- □□: Letian Cheng, Junyan Wang, Yan Gao □
- **arXiv**: □□□ 2026□2□3□
- □□: □□□□□□□□□□□□□ LLM □□□□□□□□

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6.1 □□□□□

1. □□□□□□: 1-bit□2-bit □□□□□□□□□RaBiT□ TernaryLM □□□□□□□□□□□
2. **Token** □□□□□: MoBiQuant□QTALE □□□□□ token □□□□□□□□□□□
3. □□□□□□: □□□□□□□□□□Gemma3 □ NPU □□□□□□□□□□□□□□
4. □□□□□: Astro□Quantization-Robust Unlearning □□□□□□□□□□□□□
5. □□□□□: VLA □□□Diffusion Transformer □□□□□□□

6.2 □□□□□

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1-bit □□	TernaryLM, RaBiT	□□□□	□□□□□

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□□□□□	MoBiQuant	□□□□	□□□□□
□□□□□	D²Quant, MatGPTQ	□□□□□	□□□□□□
□□□□	TOM, Gemma3 NPU	□□□□	□□□□

6.3 □□□□□□□

- □□ **1-bit** □□ □□□□□□□□□□□□
- **Token** □□□ □□□□□□□□□□□□
- **MoE** □□ □□□□□□□
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1. pQuant: Towards Effective Low-Bit Language Models via Decoupled Linear Quantization-Aware Training (arXiv:2602.23351)
2. Scaling State-Space Models on Multiple GPUs with Tensor Parallelism (arXiv:2026)
3. TOM: A Ternary Read-only Memory Accelerator for LLM-powered Edge Intelligence (arXiv:2026)
4. MoBiQuant: Mixture-of-Bits Quantization for Token-Adaptive Elastic LLMs (arXiv:2026)
5. SPQ: An Ensemble Technique for Large Language Model Compression (arXiv:2026)
6. Quecto-V1: Empirical Analysis of 8-bit Quantized Small Language Models (arXiv:2026)
7. Compact LLM Deployment and World Model Assisted Offloading in Mobile Edge Computing (arXiv:2026)
8. Quantization-Robust LLM Unlearning via Low-Rank Adaptation (arXiv:2026)
9. QTALE: Quantization-Robust Token-Adaptive Layer Execution for LLMs (arXiv:2602.22207)
10. NanoQuant: Efficient Sub-1-Bit Quantization of Large Language Models (arXiv:2026)
11. Astro: Activation-guided Structured Regularization for Outlier-Robust LLM Quantization (arXiv:2026)
12. TernaryLM: Memory-Efficient Language Modeling via Native 1-Bit Quantization (arXiv:2026)

13. MatGPTQ: Accurate and Efficient Post-Training Matryoshka Quantization (arXiv:2026)
14. D^{2}Quant: Accurate Low-bit Post-Training Weight Quantization for LLMs (arXiv:2026)
15. RaBiT: Residual-Aware Binarization Training for Accurate and Efficient LLMs (arXiv:2026)
16. Mapping Gemma3 onto an Edge Dataflow Architecture (arXiv:2026)
17. Regularized Calibration with Successive Rounding for Post-Training Quantization (arXiv:2026)
18. BPDQ: Bit-Plane Decomposition Quantization on a Variable Grid (arXiv:2026)
19. QVLA: Not All Channels Are Equal in Vision-Language-Action Model's Quantization (arXiv:2026)
20. UniComp: A Unified Evaluation of LLM Compression via Pruning, Quantization and Distillation (arXiv:2026)
21. GradPruner: Gradient-Guided Layer Pruning for LLMs (arXiv:2026)
22. Price of universality in vector quantization is at most 0.11 bit (arXiv:2026)
23. Differentially Private and Communication Efficient LLM Split Inference (arXiv:2026)
24. KBVQ-MoE: KLT-guided SVD with Bias-Corrected Vector Quantization for MoE (arXiv:2026)

□□□□ *Subagent □□□□ □□□□: 2026-03-02 07:00 (GMT+8)*